

Syslog

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 - Syslog Overview
 - Syslog Message Format
 - Syslog Facilities and Severity Levels
 - Syslog Configuration

Syslog Overview

Introduction to Syslog

Syslog is an **industry-standard protocol used for logging messages** generated by network devices. These messages record events that happen during device operation and are essential for monitoring and troubleshooting.

Devices generate Syslog messages for events such as:

1. Interface status changes (up/down)
2. Routing protocol neighbor changes
3. Configuration changes
4. System-level events

These messages can be:

1. Displayed directly in the CLI
2. Stored in memory (buffer)
3. Sent to an external Syslog server

This makes Syslog very important for tracking events and diagnosing network issues.

Example Syslog Output

```
R1(config)#int g0/0
R1(config-if)#no shutdown
R1(config-if)#
*Feb 11 03:02:55.304: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Feb 11 03:02:56.305: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

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Explanation:

When the interface is enabled using no shutdown:

1. First message → Physical interface goes up
2. Second message → Line protocol goes up

This shows how Syslog automatically logs real-time events as they occur on the device.



Syslog Message Format

Standard Format

`seq:time stamp: %facility-severity-MNEMONIC:description`

Field Breakdown

1. Sequence number (seq) → Order of messages
2. Timestamp → Time when event occurred
3. Facility → Process that generated the message
4. Severity → Importance level of the event
5. Mnemonic → Short code identifying the event
6. Description → Detailed explanation

Important Note, Sequence number and timestamp may or may not appear, depending on configuration.

Understanding Each Field (Connected View)

Each part of the Syslog message provides specific information:

1. **Sequence Number:** Helps track the order of events
2. **Timestamp:** Allows correlation of events across devices
3. **Facility:** Identifies the source (e.g., LINK, OSPF, SYS)
4. **Severity:** Indicates how serious the event is
5. **Mnemonic:** Gives a quick idea of what happened
6. **Description:** Provides full details of the event

Together, these fields make Syslog messages structured and easy to analyze.

Syslog Facilities and Severity Levels

Syslog Severity Levels

Level	Keyword	Description
0	Emergency	System is unusable
1	Alert	Action must be taken immediately
2	Critical	Critical conditions
3	Error	Error conditions
4	Warning	Warning conditions
5	Notice	Normal but significant condition (Notification)
6	Informational	Informational messages
7	Debugging	Debug-level messages

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Key Points

1. Lower number = **higher severity**
2. Level 5 is called:
 - Notice (RFC standard)
 - Notification (Cisco IOS)

Severity meaning can vary slightly between vendors.

Syslog Message Examples

Example 1

*Feb 11 03:02:55.304: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up

1. Facility → LINK
2. Severity → 3
3. Mnemonic → UPDOWN
4. Description → Interface changed state

Example 2

%OSPF-5-ADJCHG: Process 1, neighbor changed state

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1. Facility → OSPF
2. Severity → 5
3. Mnemonic → ADJCHG

Example 3 (with sequence number)

000043: *Feb 11 05:06:43.331: %SYS-5-CONFIG_I: Configured from console

1. Sequence number present
2. Facility → SYS
3. Severity → 5

Example 4

%SYS-6-CLOCKUPDATE: System clock has been updated

- Severity → 6 (Informational)

Syslog Message Destinations

Syslog messages can be sent to different locations depending on configuration:

1. Console

1. Displayed directly in CLI
2. Enabled by default
3. Shows all severity levels

2. VTY (Telnet/SSH)

Syslog messages will be displayed in the CLI when connected to the device via Telnet/SSH.

1. Displayed in remote sessions
2. Disabled by default

3. Buffer (RAM)

Syslog messages will be saved to RAM, by default all messages (Level 0-7) are displayed and we can view the messages with show logging.

4. External Syslog Server

You can configure the device to send Syslog messages to an external server.

“Syslog servers will listen for messages on UDP port 514. Remember that port number!”

Syslog Configuration Commands

Syslog Commands

Here are some basic syslog configurations.

```
!configure logging to the console line  
R1(config)#logging console 6
```

logging console *Level*
*you can use the level **number** (6) or **keyword** (informational)
*this will enable logging for the *informational* severity and higher

- Enables Syslog messages to be displayed on the console line
- Severity level 6 (informational) and all higher-priority messages (0–6) will be shown
- Messages with level 7 (debugging) will NOT be displayed

```
!configure logging to the vty lines  
R1(config)#logging monitor informational
```

logging monitor *Level*
*same points as above about the level

- Enables Syslog messages for **VTY sessions (Telnet/SSH)**
- Displays messages with severity **6 (informational)** and higher (0–6)
- Requires terminal monitor in the session to actually see the logs

```
!configure logging to the buffer  
R1(config)#logging buffered 8192 6
```

logging buffered [size] *Level*
*same points as above about the level
*buffer size is in bytes

- Stores Syslog messages in the device's **RAM buffer**
- **8192** = buffer size in bytes
- Stores messages with severity **6 and higher (0–6)**
- Logs can be viewed using: show logging
- If Buffer size is not mention it use default size

```
!configure logging to an external server  
R1(config)#logging 192.168.1.100  
R1(config)#logging host 192.168.1.100
```

logging server-ip
logging host server-ip
*these commands are the same!

- Sends Syslog messages to an **external Syslog server** at IP 192.168.1.100
- This command is equivalent to logging host

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```
R1(config)#logging trap debugging
```

logging trap *Level*
*same points as above about the level
*this sets the logging level for the external server

- Defines which severity levels are sent to the **external Syslog server**
- debugging = level **7 (lowest priority)**
- This means **all messages (0–7)** will be sent to the server

Terminal Monitor (VTY Sessions)

- Even if **logging monitor Level** is enabled, by default Syslog messages will not be displayed when connected via Telnet or SSH.
- For the messages to be displayed, you must use the following command:

```
R1#terminal monitor
```

- This command must be used **every time you connect to the device via Telnet or SSH.**

Logging Synchronous

Problem: Syslog messages interrupt CLI input while typing commands.

- By default, logging messages displayed in the CLI while you are in the middle of typing a command will result in something like this:

```
R1(config)#exit
R1#show ip in
*Feb 11 09:38:41.607: %SYS-5-CONFIG_I: Configured from console by jeremy on console
interface brief
```

- To prevent this, you should use the **logging synchronous** on the appropriate *line*. (I will talk more about 'line' configuration in the Telnet/SSH video!)

```
R1(config)#line console 0
R1(config-line)#logging synchronous
```

- This will cause a new line to be printed if your typing is interrupted by a message.

```
R1(config)#exit
R1#show ip int
*Feb 11 09:41:00.554: %SYS-5-CONFIG_I: Configured from console by jeremy on console
R1#show ip int
```

show ip int was reprinted on a new line. This makes it easier to continue typing the command.

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Timestamps and Sequence Numbers

```
R1(config)#service timestamps log ?
datetime    Timestamp with date and time
uptime      Timestamp with system uptime
<cr>
```

datetime = timestamps will display the date/time when the event occurred.
uptime = timestamps will display how long the device had been running when the event occurred.

```
R1(config)#service timestamps log datetime
R1(config)#
```

```
R1(config)#service sequence-numbers
R1(config)#exit
```

```
R1#
000039: *Feb 11 10:32:46: %SYS-5-CONFIG_I: Configured from console by
jeremy on console
```

Output Example: Sequence number and timestamp both included

Command Summary

```
R1(config)# logging console severity
R1(config)# logging monitor severity
R1(config)# logging buffered [size] severity
R1(config)# logging server-ip
R1(config)# logging host server-ip
R1(config)# logging trap severity
R1# terminal monitor
R1(config-line)# logging synchronous
R1(config)# service timestamps log [datetime | uptime]
R1(config)# service sequence-numbers
```

Syslog vs SNMP (Connected Understanding)

- Syslog and SNMP are both used for monitoring and troubleshooting of devices. They are complementary, but their functionalities are different.
- **Syslog** is used for message logging.
 - Events that occur within the system are categorized based on facility/severity and logged.
 - Used for system management, analysis, and troubleshooting.
 - Messages are sent from the devices to the server. The server **can't** actively pull information from the devices (like SNMP **Get**) or modify variables (like SNMP **Set**).
- **SNMP** is used to retrieve and organize information about the SNMP managed devices.
 - IP addresses, current interface status, temperature, CPU usage, etc.
 - SNMP servers can use **Get** to query the clients and **Set** to modify variables on the clients.